Flowchart

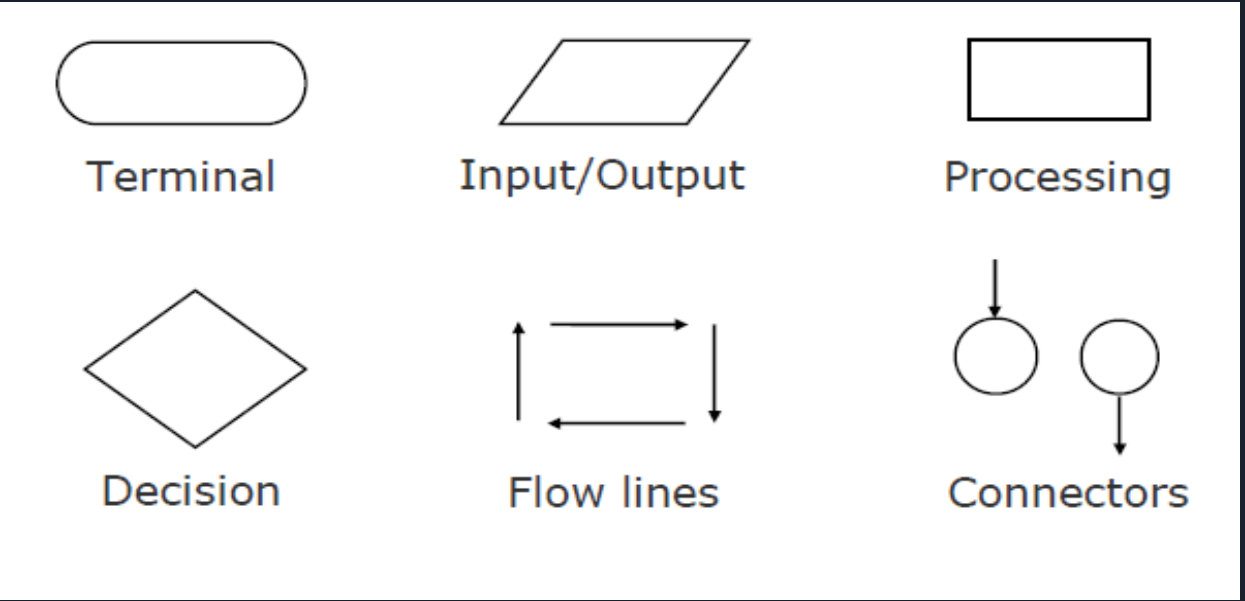
Dept. of Computer Science
Faculty of Science and Technology

Lecture No:	**04**	**Week No:**		**Semester:**	Spring 24-25
Lecturer:	***Jubayer Ahamed Roots (jubayer@aiub.edu)***				

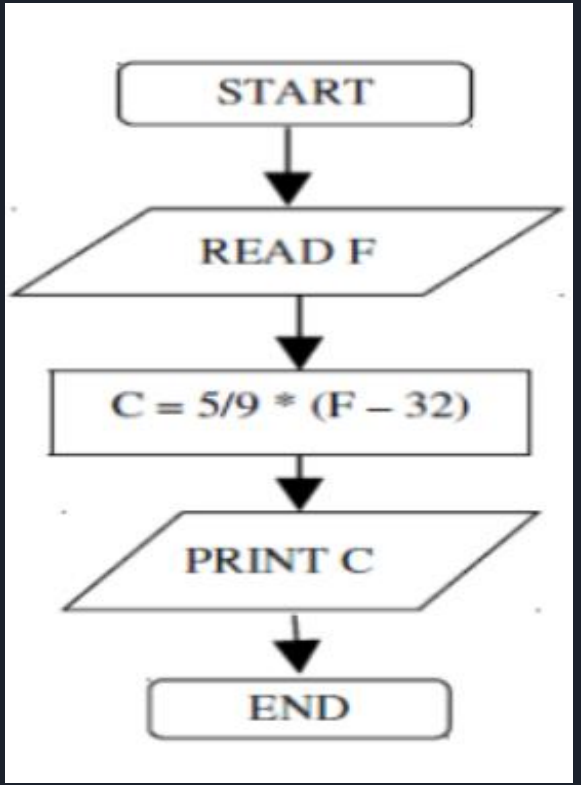
What is a flowchart?

- *Flowchart* is a pictorial representation of an algorithm
- Uses symbols (boxes of different shapes) that have standardized meanings to denote different types of instructions
- Actual instructions are written within the boxes
- Boxes are connected by solid lines having arrow marks to indicate the exact sequence in which the instructions are to be executed
- Process of drawing a flowchart for an algorithm is called *flowcharting*

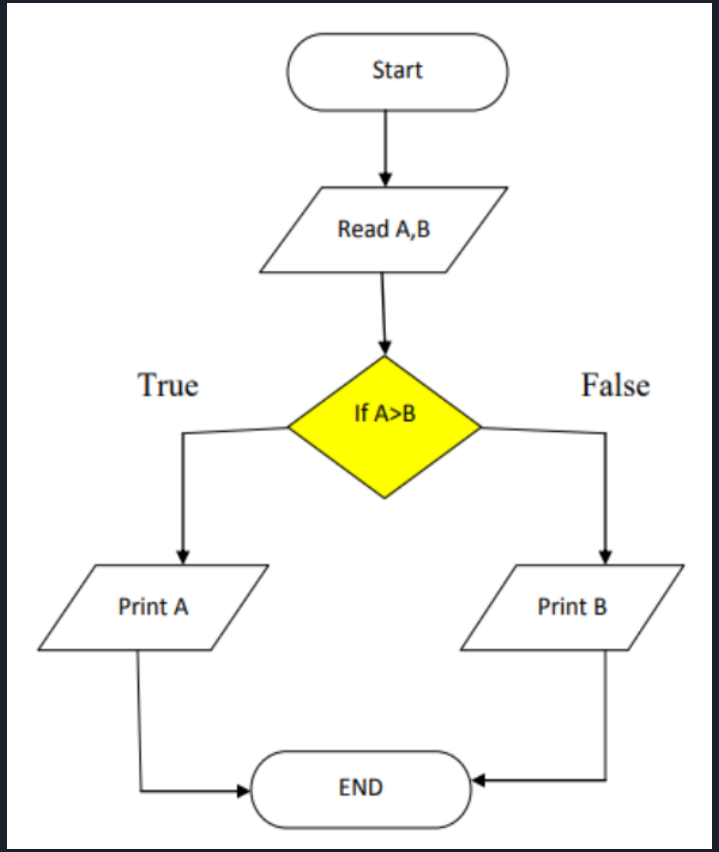
Basic flowchart symbols



Convert Fahrenheit to Celsius



Find the larger number

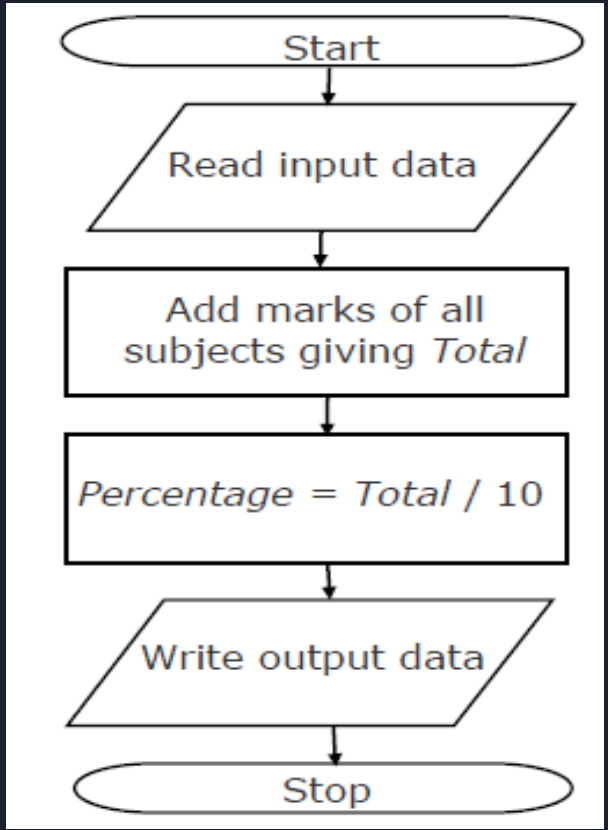


A student appears in an examination, which consists of total 10 subjects, each subject having maximum marks of 100.

The roll number of the student, his/her name, and the marks obtained by him/her in various subjects are supplied as input data.

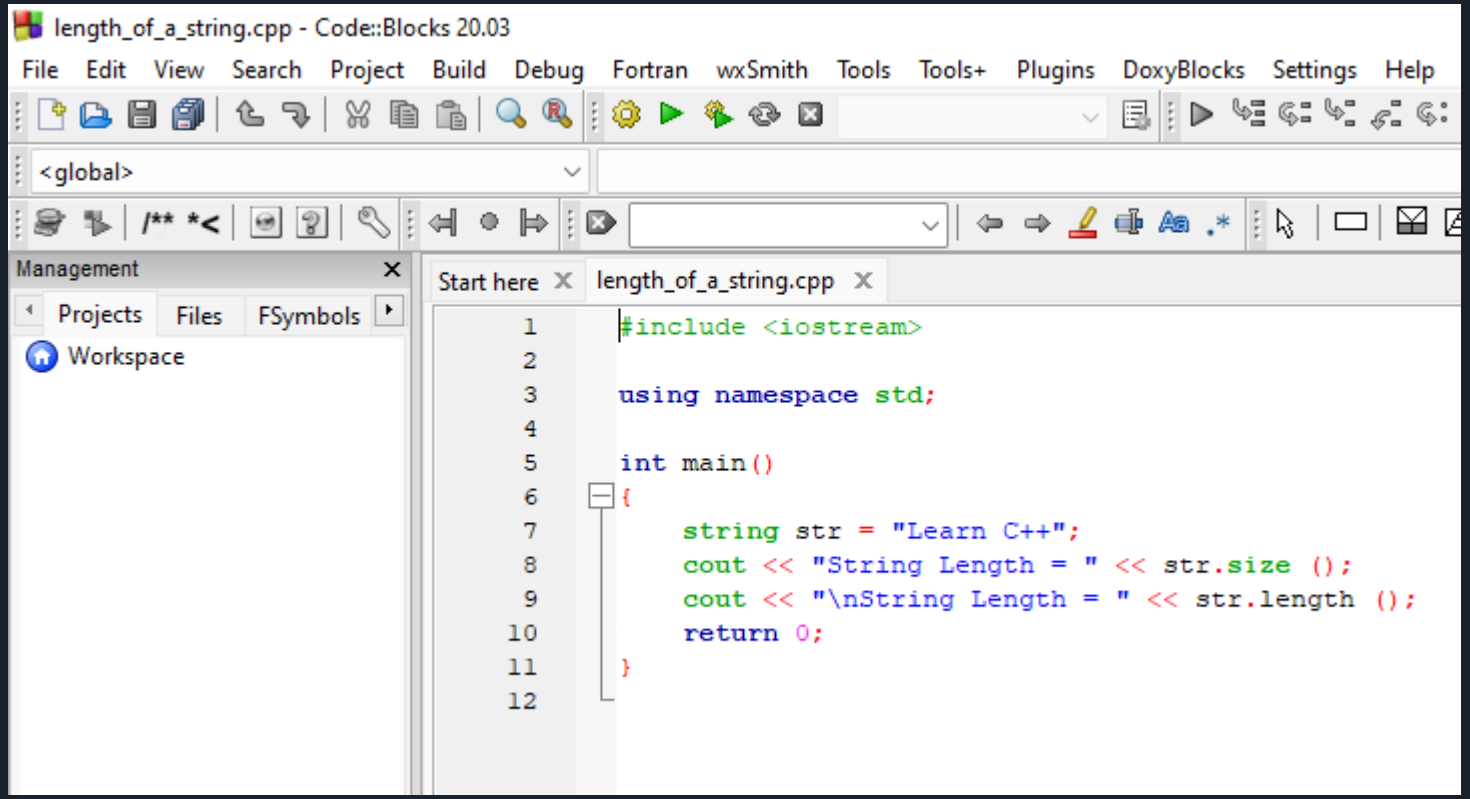
Such a collection of related data items, which is treated as a unit is known as a record.

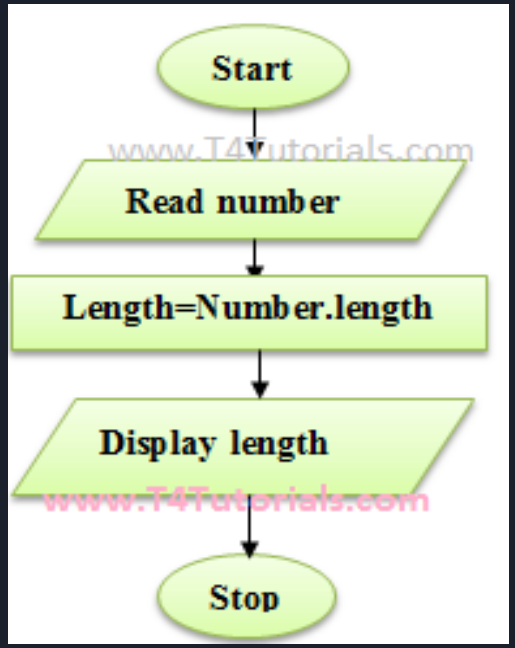
Draw a flowchart to calculate the percentage marks. Then print it along with his/her roll number and name.



Flowcharts based on the C++ Program

1. Check the length of a string





Here, Number means the string.

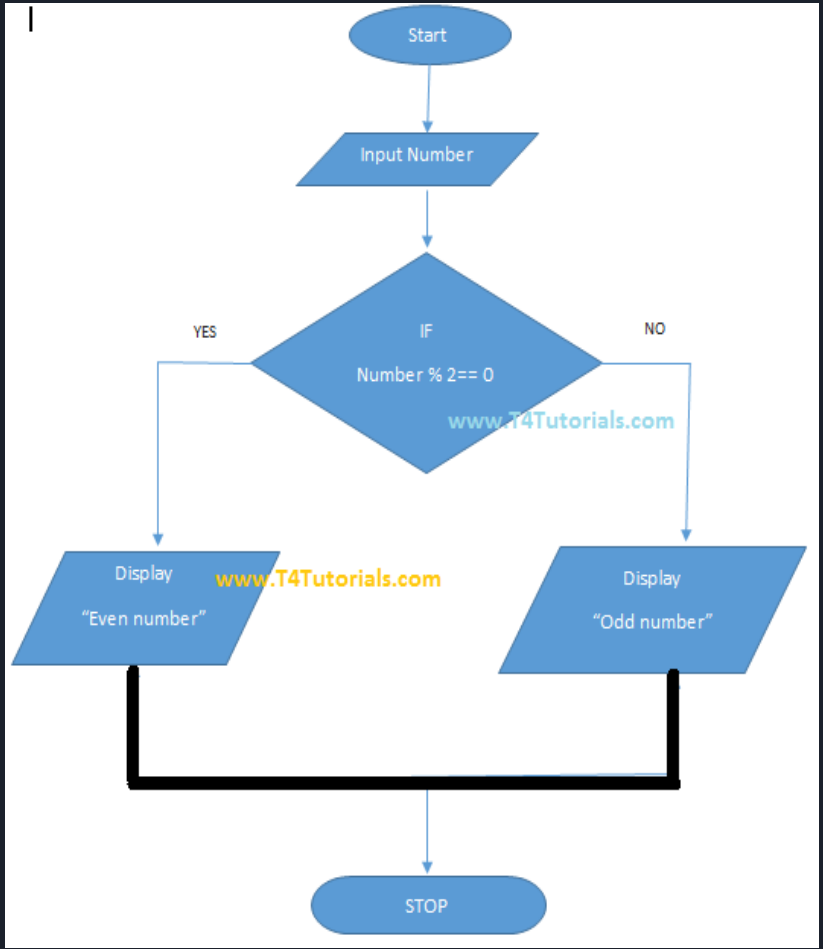
Flowchart of the length of a string

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2. Check whether a number is even or odd

```
#include<iostream>
using namespace std;
int main()
{
	int number;
	cout<<"Enter The Number You Want To Check:";
	cin>>number;
	if(number%2==0)
	{
		cout<<"Hi, Number is Even";
	}
	 else
	{
		cout<<"Hi, Number is ODD";
	}
}
```

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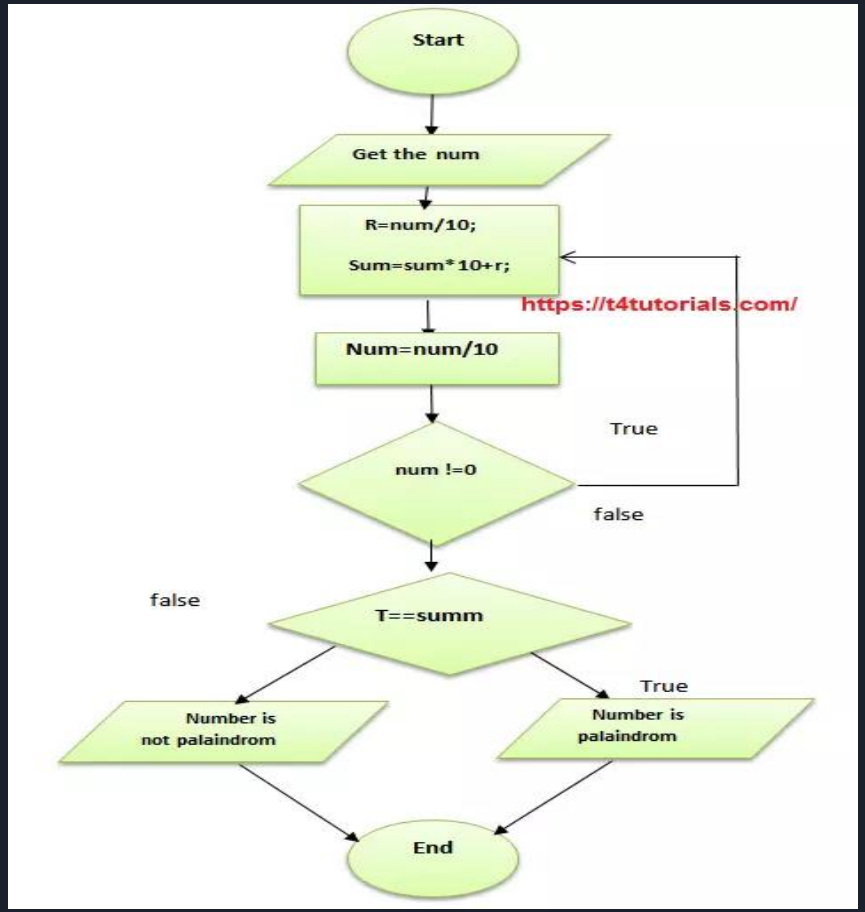


Flowchart of even-odd number program

3. Code of Palindrome Program using do-while loop

```
#include<iostream>
using namespace std;
int main()
{
	int num,r,sum=0,t;
	cout<<"enter number"<<endl;
	cin>>num;
	t=num;
	do
	{
		r=num%10;
		sum=sum*10+r;
		num=num/10;
	}
	while(num!=0);
	if(t==sum)
	{
		cout<<"The number is a palindrome:";
	}
	else
	cout<<"Number is not palidrome:";
}
```

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R = num % 10 [correction]

Flowchart of Palindrome

Tasks:

1. Find out the Max number between the two numbers. Then draw the flowchart.

2. Swap two numbers. [Ex: 1st number = 20; 2nd number = 12; Output will be: 1st number = 12 and 2nd number = 20].

3. By using this code, we can find out the area of a rectangle. Draw the flowchart by using this program.

area of rectangle.cpp

```cpp
#include <iostream>
using namespace std;

    int main()
    {
        int width, len, area;
        cout<<" Input the length of the rectangle : ";
        cin>>len;
        cout<<" Input the width of the rectangle : ";
        cin>>width;
        area=(len*width);
        cout<<" The area of the rectangle is : "<< area << endl;
        cout << endl;
        return 0;
    }
```

4. This is a code of the Fibonacci Series

```
#include <iostream>
using namespace std;

int main() {
    int n, t1 = 0, t2 = 1, nextTerm = 0;

    cout << "Enter the number of terms: ";
    cin >> n;

    cout << "Fibonacci Series: ";

    for (int i = 1; i <= n; ++i) {
        // Prints the first two terms.
        if(i == 1) {
            cout << t1 << ", ";
            continue;
        }
        if(i == 2) {
            cout << t2 << ", ";
            continue;
        }
        nextTerm = t1 + t2;
        t1 = t2;
        t2 = nextTerm;

        cout << nextTerm << ", ";
    }
    return 0;
}
```

Draw the flowchart based on this program


Thank You